Cristian Tommasino – Curriculum Vitae

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Contact

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Research group	IKNOS Lab, DIETI (led by Prof. A. M. Rinaldi)

Current position

Assistant Professor (RTD-A, fixed-term researcher), scientific sector IINF-05/A Information Processing Systems, Department of Electrical Engineering and Information Technology (DIETI), University of Naples Federico II. Start date: *to be added*.

Education

Years	Degree	Institution	Details
2024	PhD in Information and Communication Technology for Health (36th cycle)	University of Naples Federico II	Thesis: *Distilled Nuclei Segmentation for Graph-Based Whole Slide Images Analysis and Retrieval* (dated 13 December 2023; fedOA). Supervisor: Prof. A. M. Rinaldi. Defence date: *to be added*
2015–2019	MSc in Computer Engineering	University of Naples Federico II	Thesis on web document retrieval with visual and semantic techniques

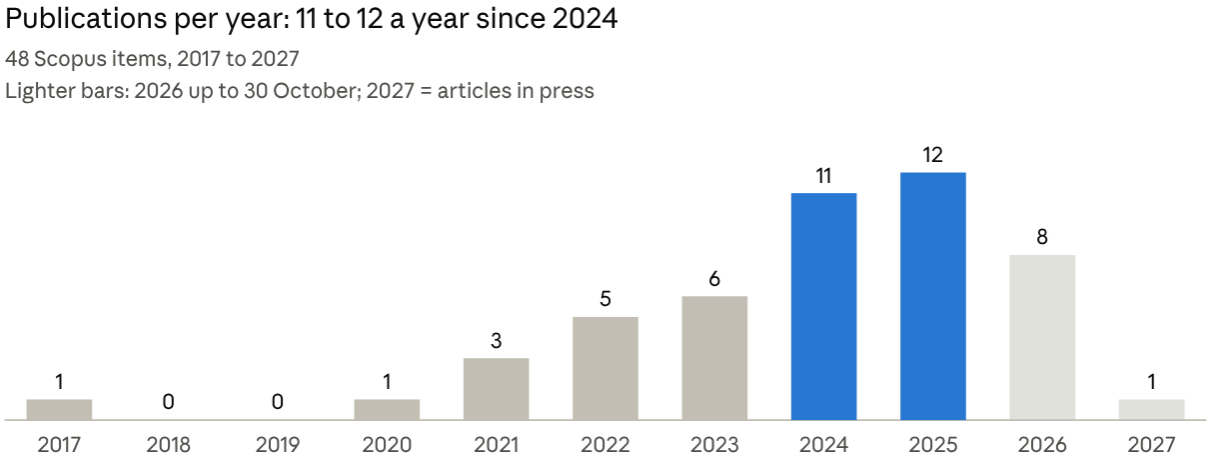
Years	Degree	Institution	Details
2011–2015	BSc in Computer Engineering	University of Naples Federico II	

Research interests

- **Computational pathology:** whole-slide image analysis, nuclei instance segmentation and classification, tissue detection, stain normalization, histopathology image retrieval.
- **Graph learning:** cell-graph construction, graph and hypergraph neural networks for cancer subtyping.
- **Generative AI and language models:** vision-language models, retrieval-augmented generation, multi-agent systems for diagnosis.
- **Knowledge representation and retrieval:** knowledge graphs, ontologies, multimedia information retrieval, focused crawling.

Research output

48 documents in Scopus (19 journal articles, 3 reviews, 25 conference papers, 1 book chapter), 308 citations, h-index 11, computed from the Scopus export of 30 October 2026. Most cited: image captioning in *Results in Engineering* (50), document classification in *Expert Systems with Applications* (28), RAG for nutrigenetics in *Procedia Computer Science* (21). Google Scholar metrics: *to be added*.



Scopus export of 30 October 2026 · 48 items

Collaborations

Partner	Topic	Evidence
Computational Pathology Group, Radboud University Medical Center, Nijmegen (F. Ciompi)	Knowledge distillation for nuclei segmentation	HoVer-UNet, IEEE ISBI 2024; code in the DIAGNijmegen repository
Pathology, University of Naples Federico II (S. Staibano, F. Merolla)	Histopathology image retrieval; biomarker inference from H&E	J. Digital Imaging 2023; WiMob 2025
Bioinformatics and oncology co-authors (N. Normanno, D. Frezzetti, F. Russo, S. Parashuraman)	LLM chatbot for RNA-seq analysis	REDAC, Bioinformatics Advances 2026
Clinical co-authors in orthofacial surgery (V. Abbate)	Generative AI for orthofacial surgery	Medicina 2025
CINI-AIIS national laboratory, Naples node	AI in healthcare	Activity reports 2023–2025 (CEUR)

Research visits: *to be added*.

Teaching

Courses listed on the UNINA teacher page:

Course	Degree programme	Academic years
Databases (Basi di Dati, 9 ECTS)	Computer Engineering (DE3)	*to be added*
Multimedia Systems (Sistemi Multimediali)	Computer Engineering (N46)	*to be added*

Service

Peer review recorded on ORCID (7 reviews):

Journal	Publisher	Reviews	Years
Medical Image Analysis	Elsevier	4	2025–2026
International Journal of Human-Computer Studies	Elsevier	1	2025
Soft Computing	Springer Nature	2	2023–2024

Other reviewing, programme committees and institutional roles: *to be added*.

Open-source software

- NuLite: lightweight model for nuclei instance segmentation and classification (BSPC 2026).
- HoVer-UNet: HoVerNet distilled into a single UNet, about three times faster at inference (ISBI 2024).

Grants, projects and awards

- Applicant: Amazon Research Awards, AWS Agentic AI call, Fall 2026 (*PathAgents*, in preparation, deadline 11 November 2026).
- Funded projects as participant, prizes and awards: *to be added*.

Publications

48 items indexed in Scopus (export of 30 October 2026), newest first within each type; own name in bold.

Journal articles (19)

1. **C. Tommasino**, C. Russo, A. M. Rinaldi. ECHO: A unified framework for efficient cell-graph construction for breast histology classification. *Biomedical Signal Processing and Control* 129, 2027. doi:10.1016/j.bspc.2026.111294
2. A. M. Rinaldi, C. Russo, **C. Tommasino**, D. Vitale. Temporal knowledge graph construction from football match videos. *Neural Computing and Applications* 38(9), 2026. doi:10.1007/s00521-026-12037-8

3. A. M. Rinaldi, C. Russo, **C. Tommasino**. Enhancing autonomous driving decision-making using knowledge graph representation and distillation techniques. *Advanced Engineering Informatics* 76, 2026. doi:10.1016/j.aei.2026.104868
4. G. M. De Filippis, P. Sahu, P. Ambrosio, S. Picascia, M. Lo Monte, I. Agliarulo, S. Di Paola, C. Russo, **C. Tommasino**, N. Normanno, D. Frezzetti, S. Parashuraman, A. M. Rinaldi, F. Russo. REDAC: RNA-seq expression data analysis chatbot. *Bioinformatics Advances* 6(1), 2026. doi:10.1093/bioadv/vbaf321
5. D. Benfenati, A. M. Rinaldi, C. Russo, **C. Tommasino**. A Generative AI Application for Qualitative Automatic Population of Multimedia Knowledge Graphs. *IEEE Transactions on Artificial Intelligence* 7(6): 3314–3323, 2026. doi:10.1109/TAI.2025.3638321
6. A. M. Rinaldi, C. Russo, **C. Tommasino**. Hierarchical graph networks for breast cancer subtype classification. *Data Mining and Knowledge Discovery* 40(6), 2026. doi:10.1007/s10618-026-01259-y
7. **C. Tommasino**, C. Russo, A. M. Rinaldi. NuLite – lightweight and fast model for nuclei instance segmentation and classification. *Biomedical Signal Processing and Control* 114, 2026. doi:10.1016/j.bspc.2025.109333
8. V. Abbate, E. Carraturo, D. Benfenati, S. Tramontano, L. A. Vaira, S. Troise, G. Canzi, G. Novelli, D. Pacella, C. Russo, **C. Tommasino**, R. Nocini, A. M. Rinaldi, G. Dell'Aversana Orabona. Text-Guided Inpainting for Aesthetic Prediction in Orthofacial Surgery: A Patient-Centered AI Approach. *Medicina (Lithuania)* 61(12), 2025. doi:10.3390/medicina61122075
9. A. M. Rinaldi, C. Russo, **C. Tommasino**. A semantic approach for cultural heritage ontology matching and integration based on textual and multimedia information. *Soft Computing* 29(2): 1019–1034, 2025. doi:10.1007/s00500-025-10517-y
10. M. Montanaro, A. M. Rinaldi, C. Russo, **C. Tommasino**. A rule-based obfuscating focused crawler in the audio retrieval domain. *Multimedia Tools and Applications* 83(9): 25231–25260, 2024. doi:10.1007/s11042-023-16155-6
11. M. Montanaro, A. M. Rinaldi, C. Russo, **C. Tommasino**. Using knowledge graphs for audio retrieval: a case study on copyright infringement detection. *World Wide Web* 27(4), 2024. doi:10.1007/s11280-024-01277-0
12. K. Madani, A. M. Rinaldi, C. Russo, **C. Tommasino**. A combined approach for improving humanoid robots autonomous cognitive capabilities. *Knowledge and Information Systems* 65(8): 3197–3221, 2023. doi:10.1007/s10115-023-01844-3
13. G. Renzi, A. M. Rinaldi, C. Russo, **C. Tommasino**. A storytelling framework based on multimedia knowledge graph using linked open data and deep neural networks. *Multimedia Tools and Applications* 82(20): 31625–31639, 2023. doi:10.1007/s11042-023-14398-x

14. A. M. Rinaldi, C. Russo, **C. Tommasino**. Automatic image captioning combining natural language processing and deep neural networks. *Results in Engineering* 18, 2023. doi:10.1016/j.rineng.2023.101107
15. **C. Tommasino**, F. Merolla, C. Russo, S. Staibano, A. M. Rinaldi. Histopathological Image Deep Feature Representation for CBIR in Smart PACS. *Journal of Digital Imaging* 36(5): 2194–2209, 2023. doi:10.1007/s10278-023-00832-x
16. M. Muscetti, A. M. Rinaldi, C. Russo, **C. Tommasino**. Multimedia ontology population through semantic analysis and hierarchical deep features extraction techniques. *Knowledge and Information Systems* 64(5): 1283–1303, 2022. doi:10.1007/s10115-022-01669-6
17. A. M. Rinaldi, C. Russo, **C. Tommasino**. An Augmented Reality CBIR System Based on Multimedia Knowledge Graph and Deep Learning Techniques in Cultural Heritage. *Computers* 11(12), 2022. doi:10.3390/computers11120172
18. A. M. Rinaldi, C. Russo, **C. Tommasino**. A semantic approach for document classification using deep neural networks and multimedia knowledge graph. *Expert Systems with Applications* 169, 2021. doi:10.1016/j.eswa.2020.114320
19. A. M. Rinaldi, **C. Tommasino**, C. Russo. A knowledge-driven multimedia retrieval system based on semantics and deep features. *Future Internet* 12(11): 1–20, 2020. doi:10.3390/fi12110183

Reviews (3)

1. G. M. De Filippis, D. Amalfitano, C. Russo, **C. Tommasino**, A. M. Rinaldi. A systematic mapping study of semantic technologies in multi-omics data integration. *Journal of Biomedical Informatics* 165, 2025. doi:10.1016/j.jbi.2025.104809
2. C. Russo, **C. Tommasino**, A. M. Rinaldi. Leveraging Graphs for Advanced Analytics in Major Team Sports: A Systematic Mapping Study. *ACM Computing Surveys* 58(6), 2025. doi:10.1145/3773024
3. D. Benfenati, D. Amalfitano, C. Russo, **C. Tommasino**, A. M. Rinaldi. Data centric Artificial Intelligence for agrifood domain: A systematic mapping study. *Computers and Electronics in Agriculture* 239, 2025. doi:10.1016/j.compag.2025.110847

Conference papers (25)

1. D. Benfenati, A. M. Rinaldi, C. Russo, **C. Tommasino**. A Generative Framework for Web Pages Classification Using Multi-modal Topic Fusion. *Communications in Computer and Information Science* 2703: 106–120, 2026. doi:10.1007/978-3-032-06878-1_6
2. D. Amalfitano, A. Abbate, D. Distante, A. M. Rinaldi, C. Russo, **C. Tommasino**. A Web Crawling-Based Process and a Graph-Based Database for Mobile Vulnerability Analysis. *Lecture Notes in Computer Science* 15749: 145–159, 2026. doi:10.1007/978-3-

031-97207-2_12

3. D. Benfenati, A. M. Rinaldi, C. Russo, **C. Tommasino**. A Multi-Sector Approach to Retrieval Augmented Generation in Agriculture. *IEEE International Conference on Systems, Man and Cybernetics (SMC)*: 5940–5945, 2025. doi:10.1109/SMC58881.2025.11343229
4. A. Moccardi, E. Cirillo, C. Davino, M. Fonisto, F. Gargiulo, R. C. Ghosh, O. Gupta, R. Jaiswal, R. L. Rovere, L. Marassi, Z. Mashaallah, N. Patwardhan, G. M. Orlando, D. Benfenati, G. M. De Filippis, A. E. Pascarella, D. Russo, C. Russo, **C. Tommasino**, S. Marrone, F. Amato, A. M. Rinaldi, V. Moscato, C. Sansone. Generative AI: Developments, Applications, and Responsible Practices. *CEUR Workshop Proceedings* 4121, 2025.
5. D. Amalfitano, D. Benfenati, D. Landolfi, A. M. Rinaldi, C. Russo, **C. Tommasino**. A Data-Driven Ontology Framework for Integrating Heterogeneous Vulnerability Sources. *CEUR Workshop Proceedings* 4176, 2025.
6. **C. Tommasino**, C. Russo, A. Crispino, S. Staibano, A. M. Rinaldi, F. Merolla. Inferring CAF-1/p60 Expression from Hematoxylin and Eosin Stained Images in Oral Squamous Cell Carcinoma. *IEEE International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*, 2025. doi:10.1109/WiMob66857.2025.11257494
7. G. M. De Filippis, A. M. Rinaldi, C. Russo, **C. Tommasino**. Advanced Topic Modeling in Genomics: Towards Personalized Dietary Recommendations Through BERTopic Analysis. *Lecture Notes in Computer Science* 15343: 3–17, 2025. doi:10.1007/978-3-031-78093-6_1
8. M. Barone, D. Benfenati, S. Capuozzo, G. M. De Filippis, A. De Simone, S. Fioretto, K. K. Georgiou, M. Gravina, J. Gortz, L. Marassi, S. Marrone, E. V. Napolitano, V. Panzetta, G. Pontillo, G. Riccio, A. Romano, C. Russo, C. Russo, **C. Tommasino**, M. Sirignano, K. Siettos, E. Masciari, A. M. Rinaldi, P. A. Netti, V. Moscato, C. Sansone. Transforming Healthcare with AI: The Work of CINI-AIIS at Federico II University. *CEUR Workshop Proceedings* 4121, 2025.
9. G. M. De Filippis, A. M. Rinaldi, C. Russo, **C. Tommasino**. Enhanced Semantic Understanding with Graph-Based Information Retrieval. *Communications in Computer and Information Science* 2197: 11–24, 2025. doi:10.1007/978-3-031-71382-8_2
10. D. Benfenati, M. Montanaro, A. M. Rinaldi, C. Russo, **C. Tommasino**. Using Focused Crawlers with Obfuscation Techniques in the Audio Retrieval Domain. *Communications in Computer and Information Science* 2022: 3–17, 2024. doi:10.1007/978-3-031-51643-6_1
11. A. M. Rinaldi, A. Romano, C. Russo, **C. Tommasino**. FPSRec: Football Players Scouting Recommendation System based on Generative AI. *IEEE International Conference on Big Data (BigData 2024)*: 7141–7150, 2024. doi:10.1109/BigData62323.2024.10825692

12. F. Amato, G. M. De Filippis, A. Galli, M. Gravina, L. Marassi, S. Marrone, E. Masciari, V. Moscato, A. M. Rinaldi, C. Russo, C. Sansone, **C. Tommasino**. Responsibile and Reliable AI: Activities of the CINI-AIIS Lab at University of Naples Federico II. *CEUR Workshop Proceedings* 3762: 101–105, 2024.
13. D. Benfenati, A. M. Rinaldi, C. Russo, **C. Tommasino**. GenCrawl: A Generative Multimedia Focused Crawler for Web Pages Classification. *IC3K 2024 – Proceedings* 1: 91–101, 2024. doi:10.5220/0012998900003838
14. F. Amato, D. Benfenati, E. Cirillo, G. M. De Filippis, M. Fonisto, A. Galli, S. Marrone, L. Marassi, V. Moscato, N. Patwardhan, A. Moccardi, A. E. Pascarella, A. M. Rinaldi, C. Russo, C. Sansone, **C. Tommasino**. Advancements and Challenges in Generative AI: Architectures, Applications, and Ethical Implications. *CEUR Workshop Proceedings* 3762: 29–34, 2024.
15. **C. Tommasino**, C. Russo, A. M. Rinaldi, F. Ciompi. HoVer-UNet: Accelerating HoVerNet with UNet-based multi-class nuclei segmentation via knowledge distillation. *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024. doi:10.1109/ISBI56570.2024.10635755
16. D. Benfenati, G. M. De Filippis, A. M. Rinaldi, C. Russo, **C. Tommasino**. A Retrieval-augmented Generation application for Question-Answering in Nutrigenetics Domain. *Procedia Computer Science* 246: 586–595, 2024. doi:10.1016/j.procs.2024.09.467
17. D. Benfenati, S. Capuozzo, G. M. De Filippis, A. De Simone, M. Gravina, L. Marassi, S. Marrone, E. Masciari, E. V. Napolitano, G. Pontillo, M. Postiglione, C. Russo, **C. Tommasino**, A. M. Rinaldi, V. Moscato, C. Sansone. AI in Medicine: Activities of the CINI-AIIS Lab at University of Naples Federico II. *CEUR Workshop Proceedings* 3762: 318–323, 2024.
18. D. Amalfitano, V. D'Angelo, A. M. Rinaldi, C. Russo, **C. Tommasino**. Enhancing Gesture Recognition for Sign Language Interpretation in Challenging Environment Conditions: A Deep Learning Approach. *IC3K 2023 – Proceedings* 1: 395–402, 2023. doi:10.5220/0012209700003598
19. A. Bosco, S. Capuozzo, B. Celano, M. Gravina, S. Marrone, M. P. Maurelli, V. Moscato, G. Pontillo, M. Postiglione, A. M. Rinaldi, L. Rinaldi, C. Russo, G. Sperlì, **C. Tommasino**, G. Cringoli, C. Sansone. AI in healthcare: Activities of the University of Naples Federico II node of the CINI-AIIS Lab. *CEUR Workshop Proceedings* 3486: 112–117, 2023.
20. A. M. Rinaldi, C. Russo, **C. Tommasino**. Effects of Color Stain Normalization in Histopathology Image Retrieval using Deep Learning. *IEEE International Symposium on Multimedia (ISM 2022)*: 26–33, 2022. doi:10.1109/ISM55400.2022.00010
21. A. M. Rinaldi, C. Russo, **C. Tommasino**. An Approach Based on Linked Open Data and Augmented Reality for Cultural Heritage Content-Based Information Retrieval. *Lecture Notes in Computer Science* 13376: 99–112, 2022. doi:10.1007/978-3-031-10450-3_8

22. A. M. Rinaldi, C. Russo, **C. Tommasino**. A Novel Approach to Populate Multimedia Knowledge Graph via Deep Learning and Semantic Analysis. *ACM International Conference Proceeding Series*: 40–47, 2022. doi:10.1145/3508397.3564846
23. A. M. Rinaldi, C. Russo, **C. Tommasino**. Visual Query Posing in Multimedia Web Document Retrieval. *IEEE 15th International Conference on Semantic Computing (ICSC 2021)*: 415–420, 2021. doi:10.1109/ICSC50631.2021.00086
24. A. M. Rinaldi, C. Russo, **C. Tommasino**. Web Document Categorization Using Knowledge Graph and Semantic Textual Topic Detection. *Lecture Notes in Computer Science* 12951: 40–51, 2021. doi:10.1007/978-3-030-86970-0_4
25. M. Ambrosino, P. Arpaia, D. Astarita, F. Bonavolontà, U. Bruno, F. Calvanese, A. Cavaliere, F. Esposito, A. Grotta, M. Marvaso, A. Scognamiglio, S. Sorrentino, **C. Tommasino**. A project of fitness telemonitoring in an information technology course. *IEEE International Workshop on Measurement and Networking (M&N 2017)*, 2017. doi:10.1109/IWMN.2017.8078358

Book chapters (1)

1. **C. Tommasino**, A. Mancuso, C. Russo, A. Masone, A. M. Rinaldi, C. Sterle, G. D. Orabona, M. Di Serafino, R. Ronza, R. La Mura, F. Verde, L. Romano. An Efficient Approach for Pancreas Segmentation in Computer Tomography Scans. *AIRO Springer Series* 12: 311–320, 2024. doi:10.1007/978-3-031-47686-0_28